

The background of the slide is a dark blue field filled with a complex, glowing network of light blue lines and dots, resembling a digital or molecular structure. The lines connect various points, creating a web-like pattern that is denser on the right side of the image.

CSAI 252 *Introduction to* *Computer* *Networks*

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El Atty

University Campus Network Design

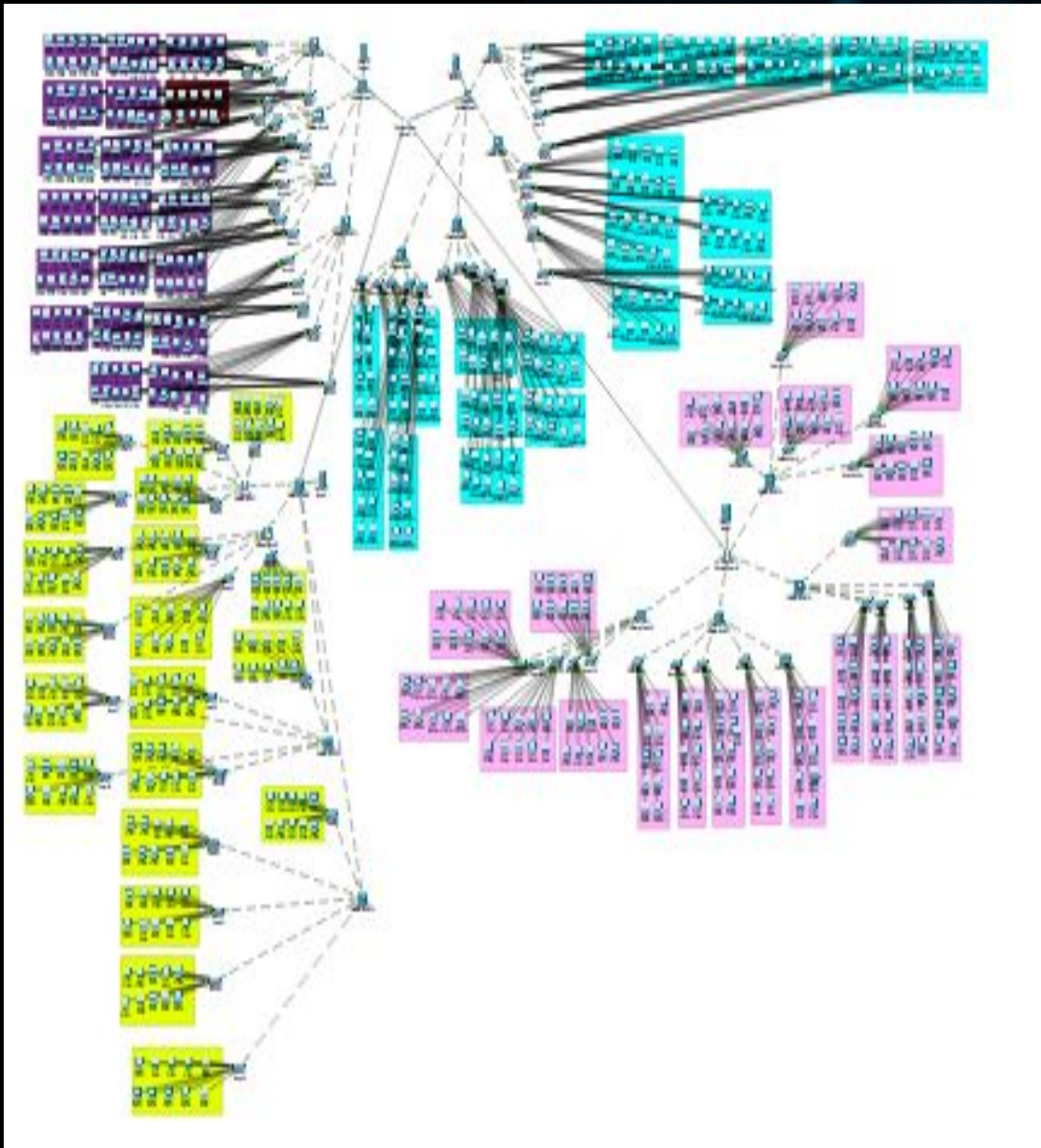
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An abstract graphic on the left side of the slide, featuring a complex network of glowing blue lines and dots, resembling a molecular structure or a data network, set against a dark blue background.

Main objective: *Design a network for small university campus*

- Design a network connecting four faculties and the Scientific Computation Center.
- Ensure connectivity between devices and sub-networks, including internet access.
- Implement Classless addressing for IP distribution.



Faculty Network Design:

- The university contain four faculty.
- In each faculty connect with Multi-switch.
- Each faculty consists of four floors.
- Each floor has five labs connected to a floor switch.
- Each lab contain at least 10 PCs
- Each faculty has 200 PCs
- The total PCs is 800 PCs
- Floor switches connect to a central faculty switch which links to the university router
- Each faculty connected with server

TOPOLOGY:

- **MESH TOPOLOGY:**

- USED within labs for redundancy and reliability.

- **STAR TOPOLOGY:**

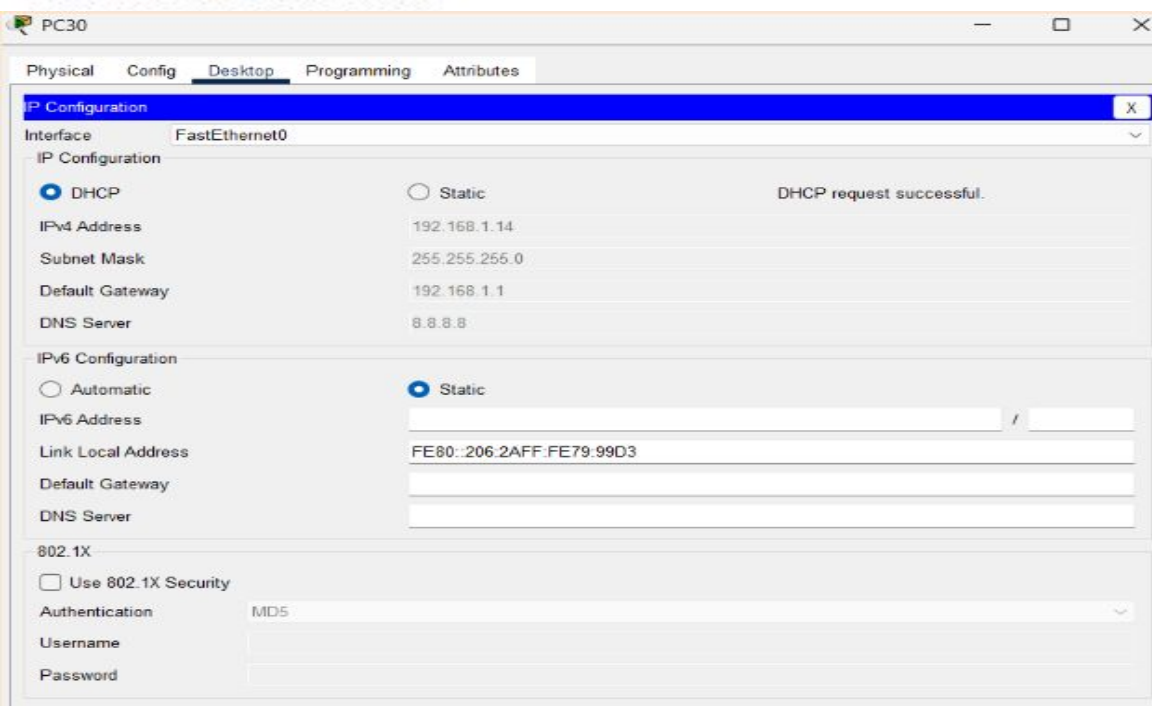
- Connected all switches on each floor to the main floor switch




```

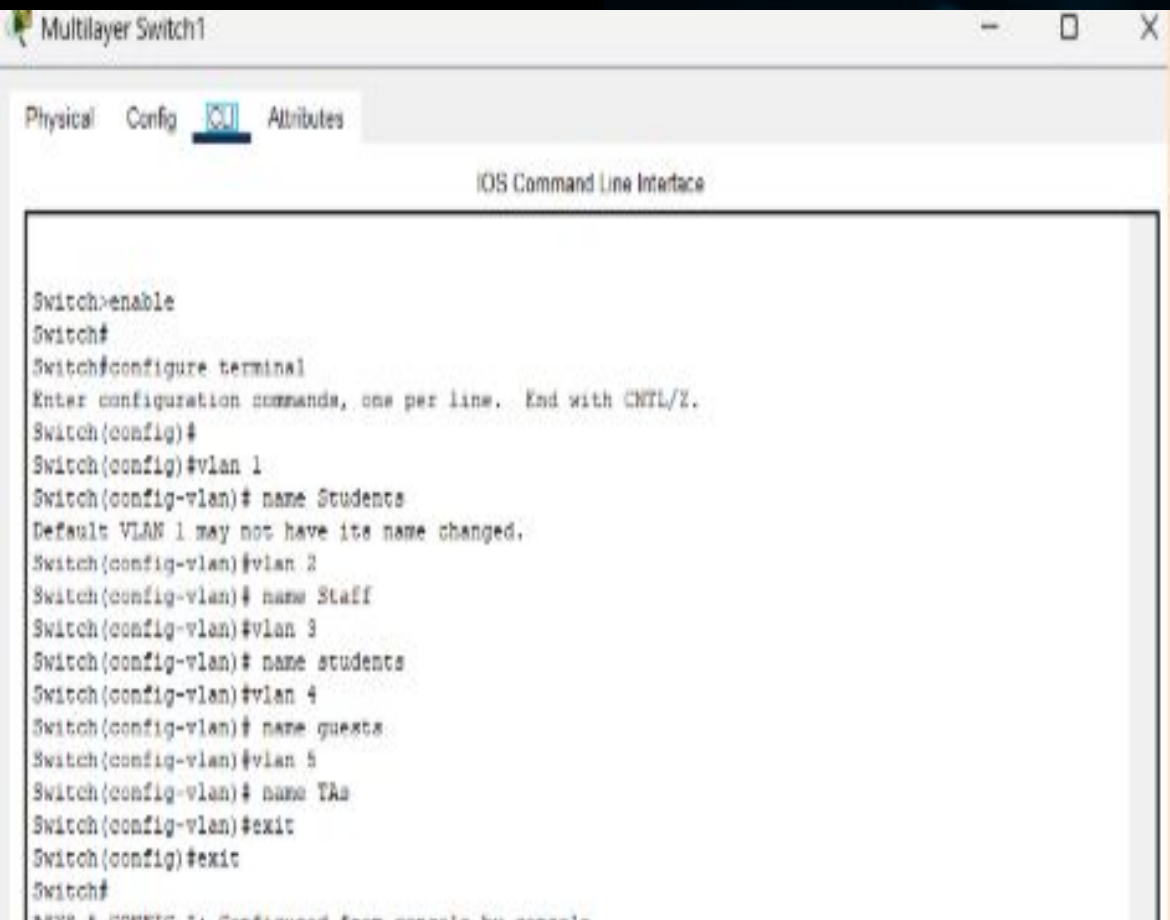
ip dhcp excluded-address 192.168.1.1 192.168.1.10
ip dhcp excluded-address 192.168.2.1 192.168.2.10
ip dhcp excluded-address 192.168.3.1 192.168.3.10
ip dhcp excluded-address 192.168.4.1 192.168.4.10
!
ip dhcp pool LAN1
 network 192.168.1.0 255.255.255.0
 default-router 192.168.1.1
 dns-server 8.8.8.8
ip dhcp pool LAN2
 network 192.168.2.0 255.255.255.0
 default-router 192.168.2.1
 dns-server 8.8.8.8
ip dhcp pool LAN3
 network 192.168.3.0 255.255.255.0
 default-router 192.168.3.1
 dns-server 8.8.8.8
ip dhcp pool LAN4
 network 192.168.4.0 255.255.255.0
 default-router 192.168.4.1
 dns-server 8.8.8.8

```



Ip address :by Dynamic Host Configuration Protocol (DHCP)

- Frist, we write the network ,ip address ,default-router and dns-server for each faculty in router command
- Each PC has its IP created by DHCP
- We converted the PC from static to DHCP
- then, ip address created dynamic



The screenshot shows a Cisco Multilayer Switch1 configuration window. The 'Config' tab is selected, and the 'IOS Command Line Interface' is visible. The terminal output shows the following commands and responses:

```
Switch>enable
Switch#
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#vlan 1
Switch(config-vlan)# name Students
Default VLAN 1 may not have its name changed.
Switch(config-vlan)#vlan 2
Switch(config-vlan)# name Staff
Switch(config-vlan)#vlan 3
Switch(config-vlan)# name students
Switch(config-vlan)#vlan 4
Switch(config-vlan)# name guests
Switch(config-vlan)#vlan 5
Switch(config-vlan)# name TAs
Switch(config-vlan)#exit
Switch(config)#exit
Switch#
```

VLAN(virtual local area network)is applied on the multi-switch of faculty.

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- VLANs implemented for segmentation by floor or department.
 - Improves security and traffic management within the network.
 - Enables logical grouping of devices even if they are on different physical networks.


```
Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-3-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-3-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 192.168.0.0
Router(config-router)#network 192.168.1.0
Router(config-router)#network 192.168.2.0
Router(config-router)#copy running-config startup-config

% Invalid input detected at '^' marker.

Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#
```

RIP(Routing Information Protocols):

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- To manage routing within the university network
 - configure the main router and switches to use RIP for routing data between faculties and the scientific computation center

summary:

- The design incorporates multiple faculties, a Scientific Computation Center, and efficient network topologies.
- VLANs and RIP enhance network security and routing efficiency.



A digital globe with a network overlay. The globe is blue and white, showing continents and oceans. It is surrounded by a complex network of glowing blue and white nodes connected by thin lines, suggesting a global network or data flow. The background is dark blue with some blurred light effects.

THANK YOU